

Single Vector Reconstruction Experiments

Results Summary — ECE 228 Final Project

Three progressive experiments test the ability to recover a single 3D magnetization vector $\mathbf{m} = (M_x, M_y, M_z)$ from scalar XMCD-style projections at multiple angles. These serve as ground-truth validation before scaling to full 2D/3D domain reconstruction.

Experiment 1 — onevector.m: Noiseless Linear Least Squares

What the system is solving: Each measurement is a dot product of the incident beam direction $\hat{\mathbf{k}}$ with the unknown vector $\mathbf{m}$, i.e. $s = \hat{\mathbf{k}} \cdot \mathbf{m}$. Sampling over a grid of (θ, ϕ) angles builds an overdetermined linear system $\mathbf{A} \cdot \mathbf{m} = \mathbf{b}$, where each row of $\mathbf{A}$ is one beam direction and $\mathbf{b}$ is the corresponding scalar measurement. The solution is the standard least-squares pseudoinverse: $\mathbf{m} = \mathbf{A} \backslash \mathbf{b}$ (MATLAB backslash), equivalent to $\mathbf{m} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$. No phase retrieval, no iteration — one direct solve.

Parameter	Value	Parameter	Value
True vector $\mathbf{m}$	[2, -1, 3]	Noise (epsilon)	0 (noiseless)
Theta range	0.2 to 1.2 rad (8 steps)	Phi range	0 to 2pi (8 steps)
Total measurements	64	System type	Overdetermined linear ($\mathbf{A} \cdot \mathbf{m} = \mathbf{b}$)
Solver	MATLAB backslash (LS pseudoinverse)	Regularization	None

Component	True Value	Recovered Value	Absolute Error	Result
M_x	2	2.0000	0.0000	✓ Exact
M_y	-1	-1.0000	0.0000	✓ Exact
M_z	3	3.0000	0.0000	✓ Exact

Result: Perfect recovery to machine precision. The sensing geometry (8x8 angle grid, rank-3 system) is sufficient and well-conditioned. Validates the forward model and linear solver.

Experiment 2 — onevectornoise.m: Regularized Least Squares (Tikhonov)

What the system is solving: Same linear system as Experiment 1 ($\mathbf{A} \cdot \mathbf{m} = \mathbf{b}$) but with optional Gaussian measurement noise (epsilon) and optional Tikhonov regularization (lambda). The regularized solution is $\mathbf{m} = (\mathbf{A}^T \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^T \mathbf{b}$, which shrinks the solution toward zero to prevent noise amplification when the system is ill-conditioned. Here both epsilon=0 and lambda=0, so this reduces to the same pseudoinverse as Experiment 1 but with a different true vector — confirming generality of the approach.

Parameter	Value	Parameter	Value
True vector $\mathbf{m}$	[9, -1, 3]	Noise (epsilon)	0 (noiseless)
Regularization (lambda)	0.0	Theta range	0.2 to 1.2 rad (8 steps)
Phi range	0 to 2pi (8 steps)	Total measurements	64
Solver	$(\mathbf{A}^T \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^T \mathbf{b}$	System type	Regularized LS

Component	True Value	Recovered Value	Absolute Error	Result
M_x	9	9.0000	0.0000	✓ Exact
M_y	-1	-1.0000	0.0000	✓ Exact
M_z	3	3.0000	0.0000	✓ Exact

Result: Perfect recovery again. Larger M_x component (9 vs 2) poses no difficulty. Confirms the linear system is robust across different vector magnitudes and directions. The regularization framework is in place for noisy experiments ($\lambda > 0$, $\epsilon > 0$).

Experiment 3 — onevecresire.m: RESIRE-V Iterative Gradient Descent

What the system is solving: Instead of a direct linear solve, RESIRE-V treats reconstruction as an optimization problem. Think of it like this: the algorithm starts with a guess for the 3D object, simulates what its scattering pattern would look like, compares that to the measured pattern, and nudges the guess in the direction that reduces the mismatch — repeating this hundreds of times. It works in two stages: (1) a scalar phase-retrieval pass that finds the object support (where magnetization is nonzero) using iterative projections, then (2) a full vector gradient descent that jointly recovers M_x, M_y, M_z by minimizing the amplitude residual between simulated and measured diffraction patterns. This is the same engine used in the full 3D pipeline (Module 6), tested here on a single voxel.

Parameter	Value	Parameter	Value
Angle bank	2 phi x 11 theta	Total projections	22
Scalar recon iters (Step 1)	100	Support voxels found	251 / 32768
Vector recon iters (Step 2)	150	Algorithm	RESIRE-V gradient descent
Optimization target	Amplitude residual $ F - A_meas $	Solver	Iterative (not direct)

Component	Percent Error	Assessment
Mx	23,040,196,621,470,744,576 %	Complete numerical divergence
My	100.00 %	Collapsed to zero (predicted 0, true != 0)
Mz	216,567,469,840,223,240,192 %	Complete numerical divergence
Overall	127,133,206,848,293,322,752 %	Total failure — NaN/Inf propagation
Vector iter error	1.271e+20 % (flat across all 150 iters)	No convergence whatsoever

Failure Mode	Likely Cause
Numerical divergence (errors $\sim 10^{20}\%$)	Forward model produces near-zero or NaN amplitudes, causing division by zero in the gradient update. The gradient explodes on the first iteration and never recovers.
Support massively overestimated (251/32768 voxels)	Scalar step found 251 nonzero support voxels in a 32^3 cube for a single-voxel object. The support constraint is far too loose, giving the optimizer too many free parameters.
Error flat across all 150 iterations	The gradient is either zero or infinite from iteration 1 — the optimizer is not receiving any useful signal. The learning rate or normalization in the gradient step is miscalibrated for this single-vector problem geometry.
My error = exactly 100%	My collapsed to zero prediction. This is the one component where 100% error is mathematically meaningful — the model predicts zero and the true value is nonzero, giving exactly 100% relative error.

Conclusion: RESIRE-V as configured completely fails on the single-vector problem. The direct linear solver (Experiments 1 & 2) trivially solves what RESIRE-V cannot. This highlights that iterative gradient methods require careful forward-model calibration, learning rate tuning, and support estimation — all of which break down at this toy scale. The full 3D pipeline failure ($L2=83.5\%$, $ang=47.8$ deg) is a milder version of the same issues.

Summary Comparison — All Three Experiments

Experiment	File	Method	True Vector	Error	Verdict
1	onevector.m	Direct LS (pseudoinverse)	[2, -1, 3]	0.000 %	✓ Perfect
2	onevectornoise.m	Regularized LS (Tikhonov, $\lambda=0$)	[9, -1, 3]	0.000 %	✓ Perfect
3	onevecresire.m	RESIRE-V (iterative gradient)	Unknown	$\sim 10^{20}$ %	✗ Diverged